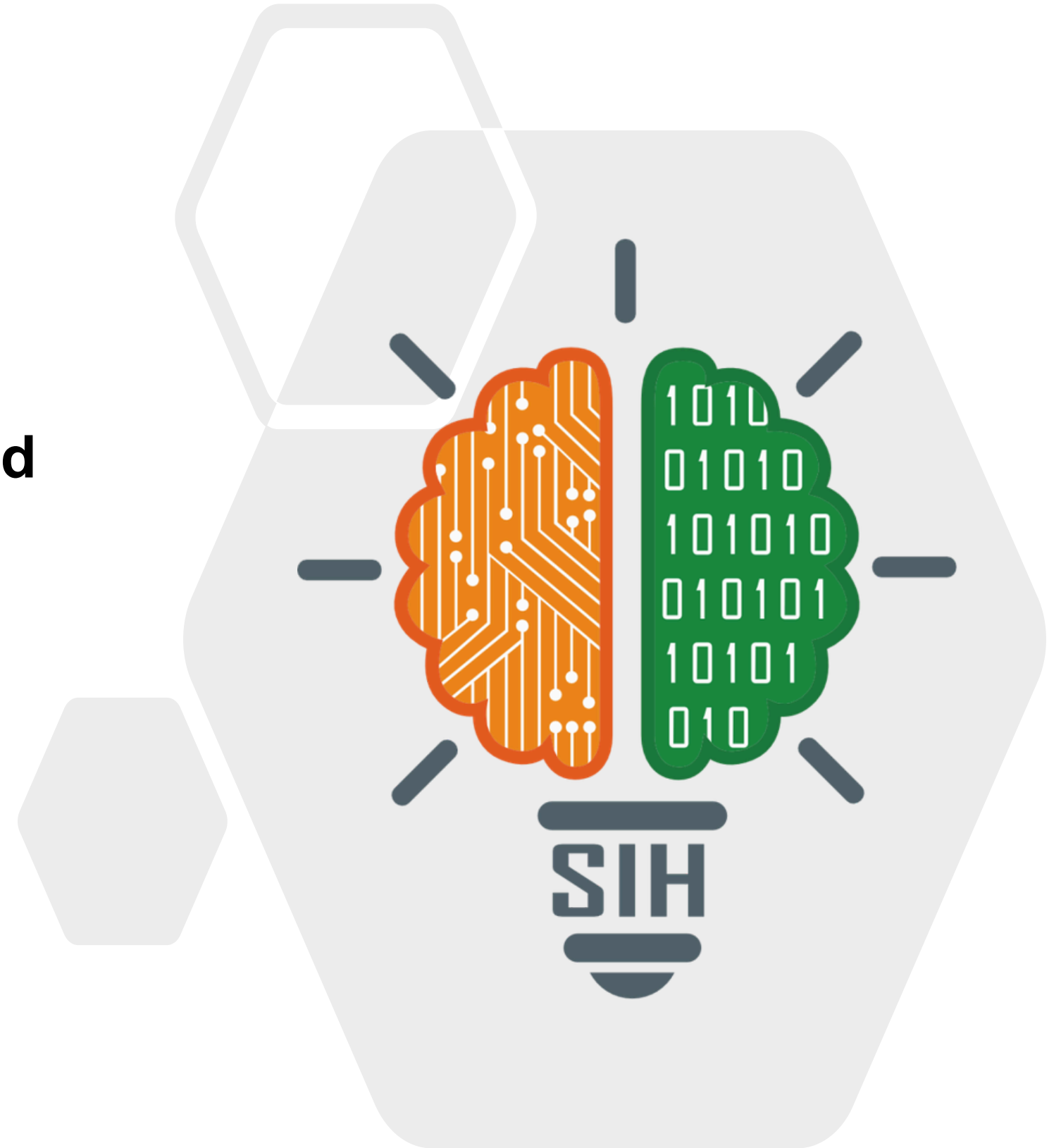


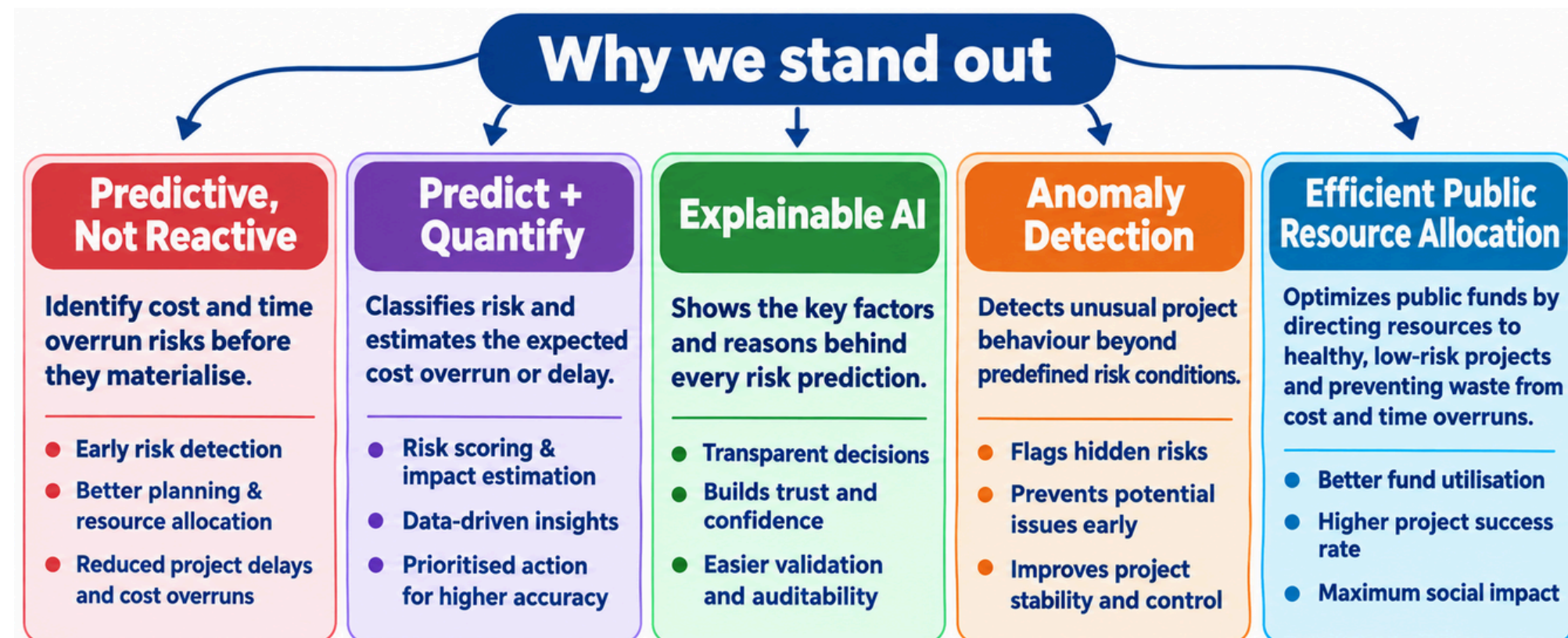
TITLE PAGE

- Problem Statement ID - 26103
- Problem Statement Title- Use case on web-based integrated project-monitoring platform
- Theme - Smart Automation
- PS Category- Software
- Team ID-
- Team Name (Registered on portal)

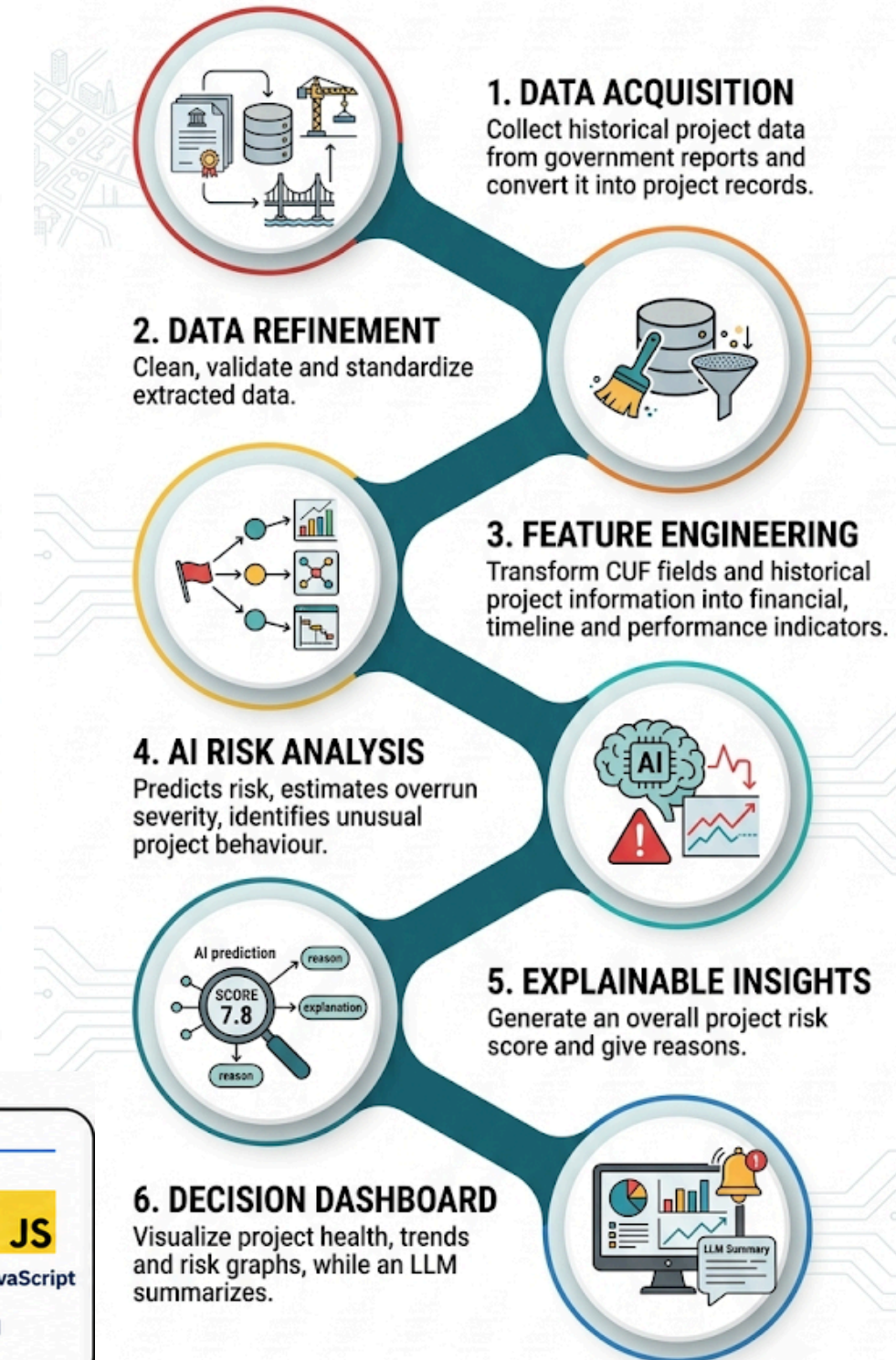
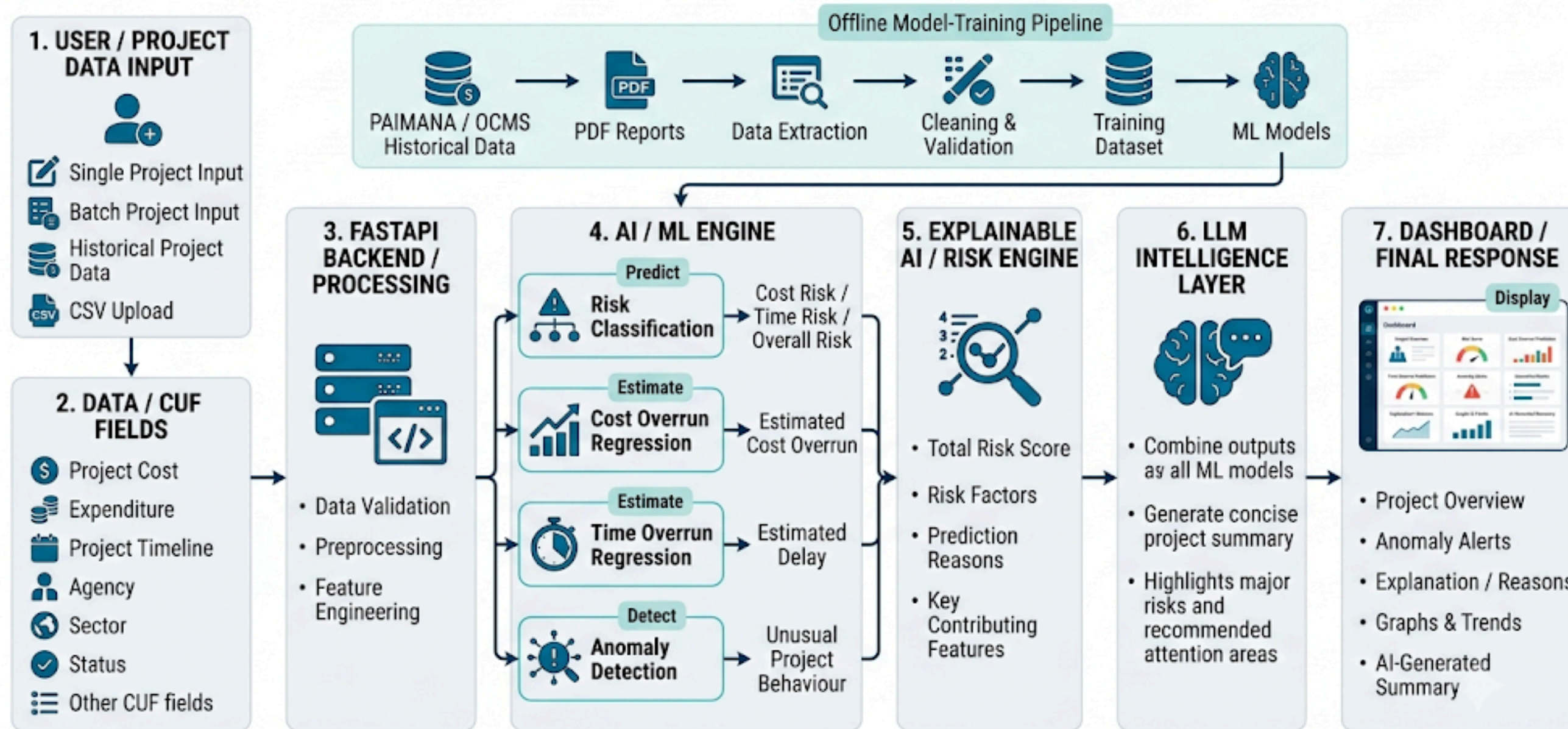


PROPOSED IDEA/SOLUTION:-

- **Predict Before Overrun** — Detect emerging cost and time risks before they become critical.
- **Quantify the Impact** — Estimate expected cost escalation and schedule delay, not just risk/no-risk.
- **Prioritize What Matters** — Rank thousands of projects so monitoring teams can focus on the highest-risk projects first.
- **Explain Every Alert** — Identify the factors driving each prediction and generate an LLM-based project summary.
- **Turn Existing Data into Decisions** — Convert PAIMANA/OCMS and CUF fields into actionable project-risk intelligence without requiring major changes to existing data collection.
- **Built for Real-World Deployment** — Supports project monitoring today and can scale toward live data ingestion, automated alerts, sector-specific models and role-based dashboards.



TECHNICAL APPROACH



My Tech Stack
Tools & Technologies

Data

Python Pandas / NumPy

Machine Learning

Scikit-learn XGBoost
XGBoost / Other Models
Classification & Regression
Anomaly Detection

Backend

FastAPI Pydantic

AI / LLM

Gemini Hugging Face
• LLM-based project summarization
• Prompt-based intelligence layer

Frontend

React HTML CSS JavaScript
• Web-based interactive dashboard
• Charts & analytical visualizations

PREDICT → EXPLAIN → PRIORITIZE → ACT



Feasibility

- ✓ **Build a strong evidence base** and identify technical and economic feasibility.
- ✓ **Map key infrastructure and resources** to be improved or developed.
- ✓ **Work with existing PAIMANA/QS technology** to reduce change burden and ensure alignment.
- ✓ **Complement with PAIMANA Analytics** to derive deeper insights and inform decisions.
- ✓ **Get buy-in from deployed units** and validate project data.
- ✓ **Evaluate from a prototype or pilot perspective** before full-scale deployment.

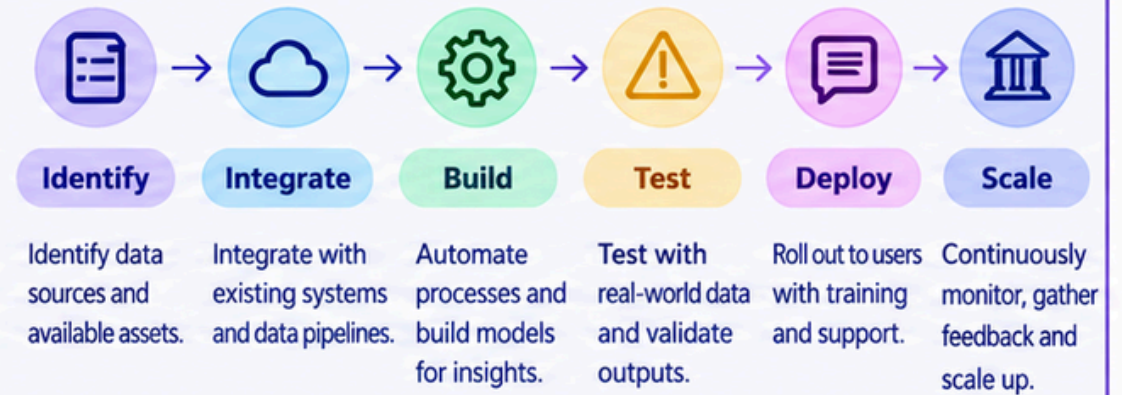


Viability

- ✓ **Addressing real and recurring problem statements** for sustainable, long-term implementation.
- ✓ **Practices can support early intervention** to address issues before they escalate.
- ✓ **Reduces dependence on manual project level analysis** and improves efficiency.
- ✓ **Enables government monitoring & reallocation** of resources for better outcomes.
- ✓ **Helps in monitoring/early detection** of potential risks.
- ✓ **Can continuously improve** with data, feedback and lessons learned over time.



Practical Implementation

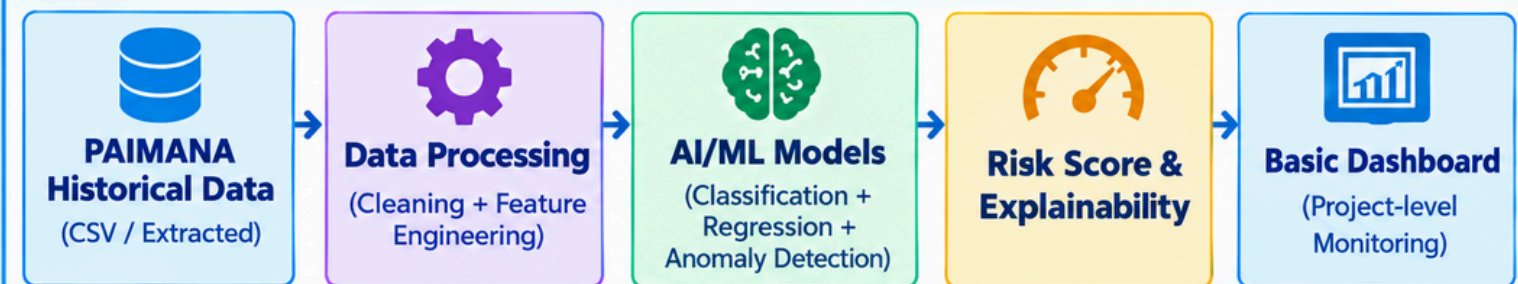


Data Acquisition Our Biggest Challenge

- 01 No Public Structured Datasets**
 - PAIMANA does not provide easily accessible public structured datasets.
- 02 Historical Data in PDF Reports**
 - Historical information is primarily available through quarterly PDF reports.
- 03 PDF Extraction Introduced**
 - We used PDF extraction to convert unstructured data into usable formats.
- 04 Data Cleaning & Transformation**
 - Extracted data required cleaning and transformation to ensure consistency and usability.
- 05 Project Tracking Across Quarters**
 - Historical project tracking required consistent project identification across quarters.
- 06 Limited Real-time Data**
 - Limited availability of live data restricted real-time ingestion.

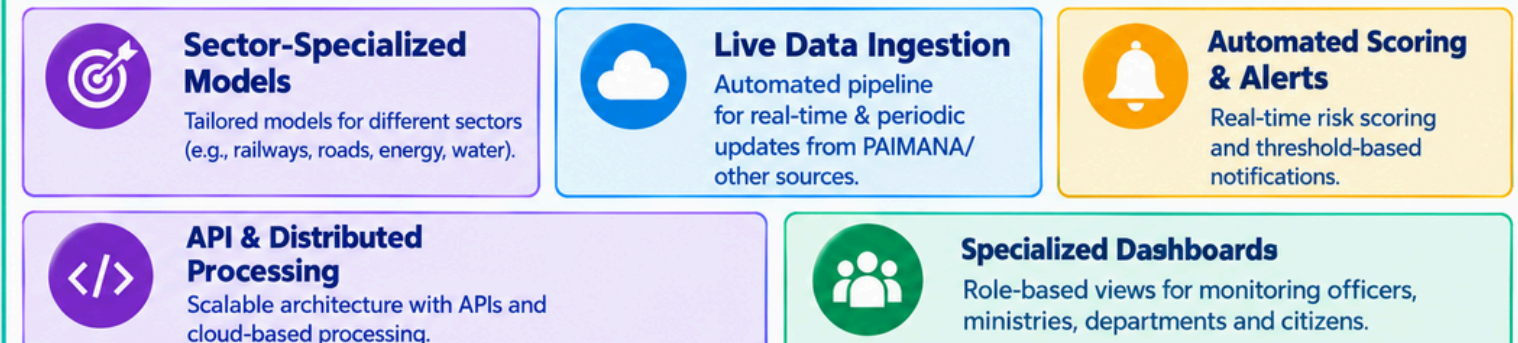


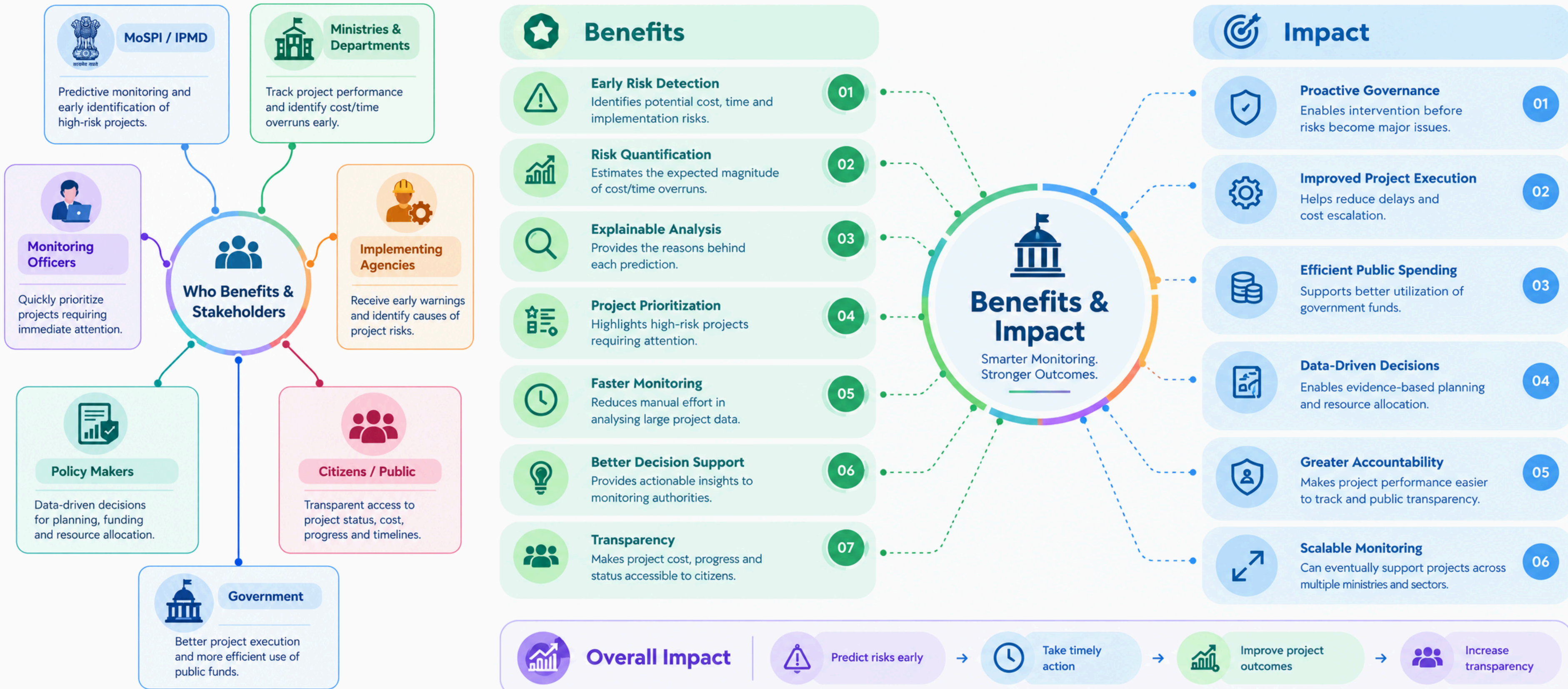
Current Prototype (SIH)



Scalability

Build for Scale | Broader Coverage | Greater Impact





1) PAIMANA Portal (MoSPI) — <https://paimana-proj.mospi.gov.in/>

Official Government of India platform for monitoring Central Sector infrastructure projects worth ₹150 crore and above; source of the PAIMANA framework itself. Our source of data.

2) Egwim et al. (2021), Machine Learning with Applications — <https://doi.org/10.1016/j.mlwa.2021.100166>

Uses applied AI methods to predict construction project delays; supports feature-based delay prediction methodology

3) <https://www.researchgate.net/publication/398083629> -

A_Hybrid_Machine_Learning_Framework_for_Electricity_Fraud_Detection_Integrating_Isolation_Forest_and_XGBoost_for_Real-World_Utility_Data. Combines Isolation Forest and XGBoost with SHAP interpretability for anomaly detection; closest methodological analogue to our hybrid anomaly detector.

Our problem statement specifically asked for a comparison between the existing conventional statistical methods and the ML approach, so we have treated the statistical models as the baseline and build our ML models based on that baseline.

Here is the complete report on the comparison and the evaluation of our models in detail. -

[https://docs.google.com/document/d/1mOF6DN4UqpZpoGFxqYhoTQ4yBYQmOGLt/edit?](https://docs.google.com/document/d/1mOF6DN4UqpZpoGFxqYhoTQ4yBYQmOGLt/edit?usp=sharing&ouid=113370278249050097271&rtpof=true&sd=true)

[usp=sharing&ouid=113370278249050097271&rtpof=true&sd=true](https://docs.google.com/document/d/1mOF6DN4UqpZpoGFxqYhoTQ4yBYQmOGLt/edit?usp=sharing&ouid=113370278249050097271&rtpof=true&sd=true)